

wheeling diode. The bulb (B1) to be switched on and off is connected between the pole of relay RL1 and neutral terminal of AC mains. It connects to the live terminal of AC mains supply via normally-opened (N/O) contacts of the relay when the relay energises.

Regulated 5V power supply for the transmitter and receiver units is derived from AC mains through step-down transformer X1, bridge rectifier (consisting of diodes D2 through D5) and regulator IC 7805 (IC2).

Assemble the transmitter and re-

ceiver circuits on two separate general-purpose PCBs and enclose in suitable cabinets. Fig. 3 shows the pin configuration of sensor TSOP1738. Mount the transmitter and receiver units on the opposite pillars of the gate as shown in Fig. 4. ●

LOW-COST HEARING AID

■ PRADEEP G.

This hearing aid circuit offers the advantages of high sensitivity, low current consumption, low

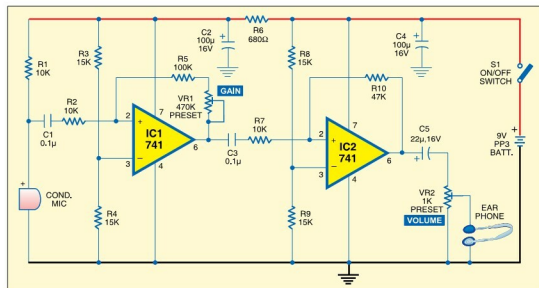
cost and light weight.

Audio signals are sensed by the condenser microphone and amplified by two op-amps built around 741 ICs. The first IC 741 (IC1) is wired as a high-

gain preamplifier, while the second IC

741 (IC2) is used as a low-gain, medium-power amplifier to drive the earphone. A 470-kilo-ohm preset (VR1) is used to control the gain, while a 1-kilo-ohm preset (VR2) is used as the output volume controller.

Assemble the circuit on a small general-purpose PCB and house it together with a 9V PP3 battery inside a cabinet the size of a cigarette pack. ●



SEQUENTIAL DEVICE CONTROL USING TV REMOTE CONTROL

■ ROHAN B. REBELLO

This circuit lets you switch on and switch off up to nine devices sequentially from your TV remote control.

The circuit consists of an IR sensor module TSOP1738, which is equivalent to TSOP1238. The output of TSOP1738 is fed to the trigger input (pin 2) of IC

NE555 (IC1) via LED1. IC1 is wired in monostable mode and its output is fed to clock input of IC CD4017 (IC2).

All the nine outputs of IC2 are connected to the bases of relay-driver transistors T1 through T9 via resistors R7 through R15, respectively. Relays RL1 through RL9 are connected between the positive terminal of 12V and collectors of respective transistors.

The circuit is powered by a 12V battery or a 12V adaptor. Since TSOP1738 needs only 5 volts (max), a 5.1V zener diode is used.

Working of the circuit is simple. At power-on, LED2 glows to indicate that the circuit is ready for operation. Now press any key on the TV remote